

Reference excerpt 01-024

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$+ \frac{1}{c} \frac{\partial w(r, z, t)}{\partial t} - \frac{w(r, z, t)}{c} = \frac{1}{c} \frac{\partial p(t, z)}{\partial z}$ (1) The radial component of the blood velocity and the non-linear terms of the equation are negligible for $c \gg w$. [Womersley(1955c)] The solution of Eq.1 and the complete procedure to calculate it are explained in detail in [Sisini et al.(2015b)]. However the solution is then defined up to the multiplicative constant c and up to the additive constant w_0 . As a consequence, the instantaneous velocity trace, calculated in this way, is proportional but not equal to the actual blood velocity.

Pressure waves propagation The pressure is transmitted along the direction AR-IJV according to the following equation $\frac{\partial p}{\partial z} = -\frac{1}{c} \frac{\partial p}{\partial t}$ (2) For each cardiac cycle, the pressure in the RA reaches its maximum value between t_P and t_Q , where t_P indicates the ECG P wave and t_Q indicates the wave Q. In this note, for the sake of simplicity, it is assumed that the pressure in the RA is maximum in correspondence of the instant t_P Q which is the average between t_P and t_Q . The relationship between the pressure in the RA and the pressure in the point G at the same instant of time is given by: $p(t + l_G/c, z_G) = p(t, z_{RA})$ (3) Where z_G is the coordinate of the US scanning point and z_{RA} is the RA coordinate. The pressure is, at the same instant t_P Q, maximum in the RA ($p(t_P$ Q, $z_{RA})$), and equal